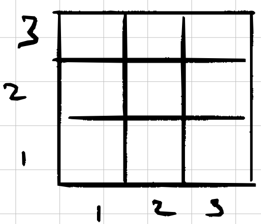
Problem 4.1

Goal: Baye's Rule → warehouse with noisy sensors

Setup: Robot position = (1, 1)
3×3 grid



Define Random variables (considering 2, 1 only)
- D: floor at (2,1) is damaged or not. (D=1) or (D=0)
- C: Detect creaking at (1,1) or not. (C=1) or (C=0)

Prior:
- $P(D=1) = 0.15$ 15% of floor damaged

- Sensor model
  $P(C=1 \mid D=1) = 0.85$ (Given true positive)
  $P(C=1 \mid D=0) = 0.08$ (Given false positive)

1) No creaking observed. Compute posterior probability that (2,1) has damaged floor.

aka what is $P(D=1 \mid C=0)$?

step 1: what are the likelihood of C=0
$P(C=0 \mid D=1) = 1 - 0.85 = 0.15$ (false negative)
$P(C=0 \mid D=0) = 1 - 0.08 = 0.92$ (true negative)

step 2: Compute $P(C=0)$ via law of probability
$$P(C=0) = P(C=0 \mid D=1)P(D=1) + P(C=0 \mid D=0)P(D=0)$$
$$= (0.15)(0.15) + (0.92)(1 - 0.15)$$
$$= 0.8045$$

step 3: Apply Baye's Rule
$$P(D=1 \mid C=0) = \frac{P(C=0 \mid D=1)P(D=1)}{P(C=0)}$$
$$= \frac{(0.15)(0.15)}{0.8045}$$
$$\simeq 0.0280$$

No creaking reduces the probability of damage from 15% to about 2.8%

2) Creaking observed at (1,1). Compute the posterior probability that $P(D=1)$ at (2,1)

aka what is $P(D=1 \mid C=1)$?

step 1: All likelihoods that $C=1$

$$P(C=1 \mid D=1) = 0.85$$
$$P(C=1 \mid D=0) = 0.08$$

step 2: Compute $P(C=1)$ via law of Probability

$$P(C=1) = P(C=1 \mid D=1)\,P(D=1) + P(C=1 \mid D=0)\,P(D=0)$$
$$= (0.85)(0.15) + (0.08)(1 - 0.15)$$
$$= 0.1955$$

step 3: Apply Bayes's Rule

$$P(D=1 \mid C=1) = \frac{P(C=1 \mid D=1)\,P(D=1)}{P(C=1)}$$
$$= \frac{(0.85)(0.15)}{0.1955}$$
$$\approx 0.6522$$

Hearing creaking raises the probability from 15% to 65%

Check: The two posteriors should be consistent with the evidence probabilities:

$$P(D=1 \mid C=0) \times P(C=0) + P(D=1 \mid C=1) \times P(D=1)$$
$$= P(D=1)$$

$$(0.0280)(0.8045) + (0.6522)(0.1955)$$
$$\approx 0.0225 + 0.1275 = 0.15 \checkmark$$

3) Sensor Comparison

New sensor sensor model

$P_N(C=1 \mid D=1) = 0.95$ (true positive)

$P_N(C=1 \mid D=0) = 0.02$ (false positive)

3-1) No cracking. Compute $P_N(D=1 \mid C=0)$

Step 1: likelihood of $C=0$

$P_N(C=0 \mid D=1) = 1 - 0.95 = 0.05$

$P_N(C=0 \mid D=0) = 1 - 0.02 = 0.98$

Step 2: Compute $P(C=0)$ via Law of Probability

$$P_N(C=0) = P_N(C=0 \mid D=1)P_N(D=1) + P_N(C=0 \mid D=0)P_N(D=0)$$

$$= (0.05)(0.15) + (0.98)(1-0.15)$$

$$= 0.8405$$

Step 3: Apply Baye's Rule

$$P_N(D=1 \mid C=0) = \frac{P_N(C=0 \mid D=1)P_N(D=1)}{P_N(C=0)}$$

$$= \frac{(0.05)(0.15)}{0.8405}$$

$$\approx 0.0089$$

2.1) Creaking at (1,1). Compute posterior probability that $P(D=1)$ at (2,1)

$$P_N(D=1 \mid C=1)\,?$$

Step 1: All likelihood of $C=1$

$P_N(C=1 \mid D=1) = 0.95$ T.P.

$P_N(C=1 \mid D=0) = 0.02$ F.P.

Step 2: Compute $P_N(C=1)$ via Law of Pobability

$$P_N(C=1) = P_N(C=1 \mid D=1)P_N(D=1) + P_N(C=1 \mid D=0)P_N(D=0)$$
$$= (0.95)(0.15) + (0.02)(1-0.15)$$
$$= 0.1595$$

Step 3: Apply Baye's Rule

$$P_N(D=1 \mid C=1) = \frac{P_N(C=1 \mid D=1)\,P(D=1)}{P_N(C=1)}$$
$$= \frac{(0.95)(0.15)}{0.1595}$$
$$\approx 0.8934$$

Compared to 0.6522, this is a 36.985% improvement.

4. Multiple adjacent squares

Define :

- D_1 : Damage at $(2,1)$, D_2 : Damage at $(1,2)$ both with prior 0.15
- D_1, D_2 are independent
- A : at least one adjacent square is damaged

Step 1 : Pobability of at least one damaged square

$$P(A) = 1 - P(D_1 = 0)\,P(D_2 = 0)$$
$$= 1 - (0.85)(0.85) = 0.2775$$

So $P(\text{none damaged}) = 0.7225$

Step 2 : Compute $P(C=0)$ via total probability

$$P(C=0) = P(C=0 \mid A)\,P(A) + P(C=0 \mid \bar{A})\,P(\bar{A})$$
$$= (1-0.85) \times 0.2775 + (1-0.08) \times 0.7225$$
$$= 0.7063$$

Step 3 : Compute $P(D_1=1 \mid C=0)$

We need $P(D_1=1, C=0)$ which requires summing over D_2 :

$$P(D_1=1, C=0) = \sum_{d_2} P(C=0 \mid D_1=1, D_2=d_2)\,P(D_1=1) \cdot P(D_2=d_2)$$

when $D_1 = 1$, at least one square is damaged regardless of D_2 so

$$P(C=0 \mid D_1=1, D_2=d_2) = 1 - 0.85 = 0.15$$

for both values of d_2

$$P(D_1=1, C=0) = 0.15 \times 0.15 \times 0.85 + 0.15 \times 0.15 \times 0.15$$
$$= 0.0225$$

$$\therefore \quad P(D_1=1 \mid C=0) = \frac{0.0225}{0.7063} \approx 0.0319$$

Problem 4.2

Setup : 4-rooms - A, B, C, D
Prior: $b_0(x) = P(\text{Package in } x) = 0.25$ for each x
Scanner: $P(\text{beep} \mid \text{package}) = 0.9$ (T.P)
$P(\text{beep} \mid \text{no package}) = 0.05$ (F.P.)

1) Room A, no beep
Enters room and no beep
Let $E_1 = \overline{\text{beep}}$ denote this evidence
Update the belief for each hypothesis about where the package is

Likelihoods of "no beep in Room A" under each hypothesis:

scanner missed package

- Package is in A : $P(E_1 \mid A) = 1 - 0.9 = 0.1$ (F.N)
- Package is in B : $P(E_1 \mid B) = 1 - 0.05 = 0.95$ (T.N)
- Package is in C : $P(E_1 \mid C) = 0.95$ (same as above)
- Package is in D : $P(E_1 \mid D) = 0.95$ "

Unnormalized posteriors $P(E_1 \mid x)\, b_0(x)$:

Room X	$P(E_1 \mid X)$	$b_0(X)$	Unnormalized
A	0.1	0.25	0.025
B	0.95	0.25	0.2375
C	0.95	0.25	0.2375
D	0.95	0.25	0.2375

Normalizing constant : $Z = 0.025 + 3(0.2375)$
$= 0.025 + 0.7125 = 0.7375$

Normalized posteriors $b_1(X)$:

$$b_1(A) = \frac{0.025}{0.7375} \approx 0.0339$$

$$b_1(B) = b_1(C) = b_1(D) = \frac{0.2375}{0.7375} \approx 0.3220$$

check : $0.0339 + 3(0.3220) = 1$ ✓

The no-beep in room A dramatically reduced b(A) from 25% to 3.4%, and the probability mass shifted roughly equally to the other three rooms.

2) Room B, Beep

Robot enters room B and beeps.

Let E_2 = beep

Likelihoods of "beep in room B" under each hypothesis

- Package in A : $P(E_2 | A) = 0.05$ (no package in B. F.P.)
- Package in B : $P(E_2 | B) = 0.9$ (package in B. T.P.)
- Package in C : $P(E_2 | C) = 0.05$ (package is in C, not B. FP.)
- Package in D : $P(E_2 | D) = 0.05$ "

Unnormalized posteriors $P(E_2 | X)\, b_1(X)$

Room X	$P(E_2 \mid X)$	$b_1(X)$	Unnormalized
A	0.05	0.0339	0.001695
B	0.90	0.3220	0.28983
C	0.05	0.3220	0.01610
D	0.05	0.3220	0.01610

Normalizing constant : $Z = 0.001695 + 0.28983 + 2(0.01610)$

$= 0.32373$

Normalized posteriors $b_2(X)$:

$$b_2(A) = \frac{0.001695}{0.32373} \approx 0.0052$$

$$b_2(B) = \frac{0.28983}{0.32373} \approx 0.8953$$

$$b_2(C) = b_2(D) = \frac{0.01610}{0.32373} \approx 0.0497$$

check: $0.0052 + 0.8953 + 2(0.0497) = 1$ ✓

After two observations, the robot is about 90% confident the package is in Room B. Room A has been nearly eliminated (0.5%) and Room C and D each have about 5%.

3) Rooms to 99% confidence (Worst Case)

Key observations: With noisy sensors, no single observation can give clarity. Each no-beep reduces a room's posterior but does not eliminate it. Each beep boosts a room's posterior but doesn't confirm it. The worst case for the number of visits occurs when the package is in the last room the robot checks, so the robot accumulates only no-beep evidence for the first several visits

Scenario: The robot visits A, B, C, D in this order. Package is in D. Assume the robot gets the most likely observation at each step (no-beep when the package is absent, beep when present).

After visiting A (no-beep), from question (1) above, $b(A) \approx 0.034$, $b(B) = b(C) = b(D) \approx 0.322$

After visiting B (no beep), the likelihood of no-beep is 0.01 if the package is in B (false negative) and 0.95 if the package is elsewhere. So B's posterior drops sharply, while all other rooms' posterior increase slightly after normalization.

- Unnormalized: A : $0.095 \times 0.034 = 0.032$
  B : $0.10 \times 0.322 = 0.032$
  C :

- Normalizing ($Z = 0.566$)
  $b(A) \approx 0.080$
  $b(B) \approx 0.080$
  $b(C) \approx 0.080$
  $b(D) \approx 0.760$

After 3 visits (all no-beep), the robot is 76% confident the package is in D - the only unvisited room. Not yet 99%

After visiting D (beep)

- Unnormalized: A : $0.05 \times 0.080 = 0.0040$
  B : $0.05 \times 0.080 = 0.0040$
  C : $0.05 \times 0.080 = 0.0040$
  D : $0.9 \times 0.760 = 0.6840$

- Normalizing ($Z = 0.696$)
  $b(D) \approx 0.983$. Close but not yet 99%

Answer: 5 visits (A, B, C, D, D again). Visiting all 4 rooms once yields ≈ 98% confidence; one additional visit to the room that beeped pushes past 99%. With noisy sensors, repeated evidence is need to reach high confidence — a single beep is not enough because it could be a false positive.

4) Perfect Scanner

With $P(\text{beep} \mid \text{package here}) = 1$ and $P(\text{beep} \mid \text{no package}) = 0$

No beep in Room A : $P(\text{no beep} \mid \text{package in } A) = 0$ so the likelihood is zero after normalization.

$$b_1(A) = 0, \quad b_1(B) = b_1(C) = b_1(D) = 1/3$$

The package is definitely not in A. One observation eliminates a room entirely.

Beep in Room B : $P(\text{beep} \mid \text{package in } B) = 1$ and $P(\text{beep} \mid \text{package elsewhere}) = 0$

So

$$b_2(B) = 1, \quad b_2(\text{others}) = 0$$

One beep identifies the room with certainty

Connection to the logical agent (Chapter 3): With a perfect scanner, the Bayesian update reduces to logical elimination. A no-beep entails that the package is not in the current room (the hypothesis is ruled out with probability 0, not merely reduced). A beep entails that the package is here (probability 1). This is exactly how the knowledge-based agent from Chapter 3 operates: each percept narrows the set of possible worlds until only one remains. The probabilistic framework subsumes the logical one as a special case - when sensors are perfect, posterior collapse to 0 or 1, and Bayesian reasoning becomes logical deduction.

Problem 4.3 - Formulating an MDP

Grid reference
Using (x, y) coordinates with $(1,1)$ at bottom-left:

(1,3)	(2,3)	(3,3)	(4,3)	- row 3 (top)
(1,2)	(2,2)	(3,2)	(4,2)	- row 2 (middle)
(1,1)	(2,1)	(3,1)	(4,1)	- row 1 (bottom)

- (4,3) : goal, reward +1, terminal
- (4,2) : hazard, reward −1, terminal
- (2,2) : wall (not a valid state)
- All other cells : reward −0.04
- Start : (1,1)

1) State and Action spaces
States : All grid cells except the wall at (2,2):

$$S = \{(1,1), (2,1), (3,1), (4,1), (1,2), (3,2), (4,2), (1,3), (2,3), (3,3), (4,3)\}$$

That is 11 states total:

Terminal states : (4,3) (goal) and (4,2) (hazard)
Non-terminal states : $11 - 2 = 9$
Actions : $A = \{N, S, E, W\}$

2) Transition Probabilities from (1,2) Taking East
From (1,2), the intended action is East (toward (2,2))

- Intended (East, $p = 0.8$): (2,2) is a wall → Stay @ (1,2)
- Drift left (North, $p = 0.1$): (1,3) is a valid cell → move to (1,3)
- Drift Right (South, $p = 0.1$): (1,1) is a valid cell → move to (1,1)

Since the intended move is blocked, that probability stays at the current cell:

$T((1,2) \mid (1,2), East) = 0.8$ (Intended blocked by wall)
$T((1,3) \mid (1,2), East) = 0.1$ (drift North)
$T((1,1) \mid (1,2), East) = 0.1$ (drift South)

Check: $0.8 + 0.1 + 0.1 = 1$ ✓

The wall at (2,2) makes east from (1,2) essentially a "stay" action with small chances of drifting north or south.